

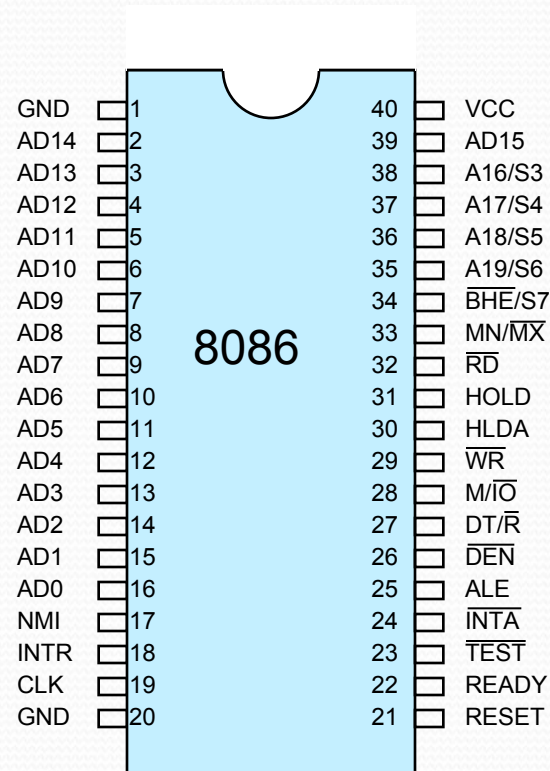
Microprocessors and Microcontrollers CSE – 3103

Course Teacher: Md. Shovon

Basic Features

- 8086 announced in 1978; 8086 is a 16 bit microprocessor with a 16 bit data bus
- 8088 announced in 1979; 8088 is a 16 bit microprocessor with an 8 bit data bus
- Both manufactured using High-performance Metal Oxide Semiconductor (HMOS) technology
- Both contain about 29000 transistors
- Both are packaged in 40 pin dual-in-line package (DIP)

8086/8088 Pinout Diagrams



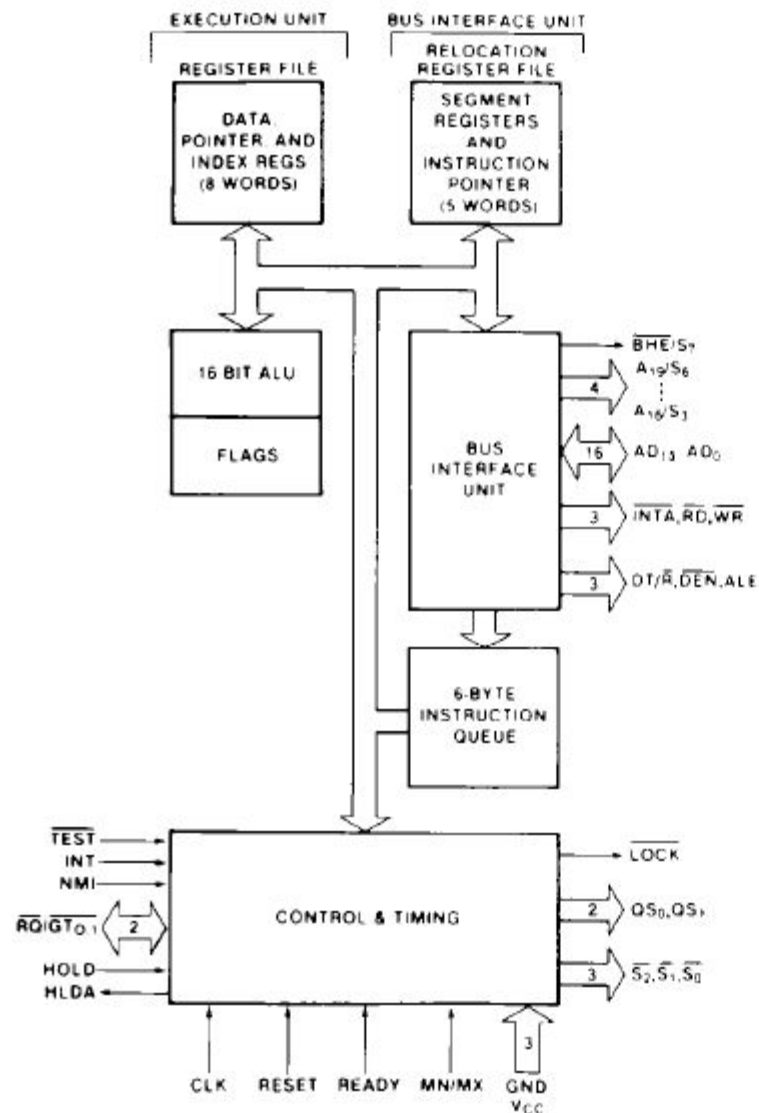
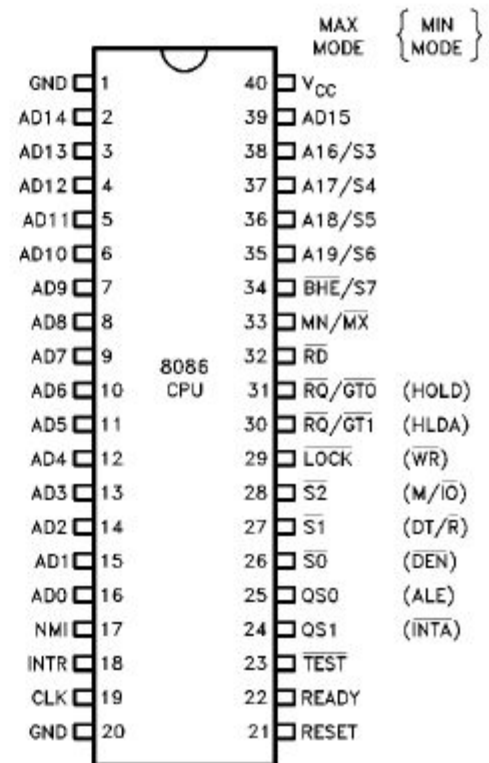


Figure 1. 8086 CPU Block Diagram

231455-1



231455-2

40 Lead

Figure 2. 8086 Pin Configuration

Minimum-mode and Maximum-mode Systems

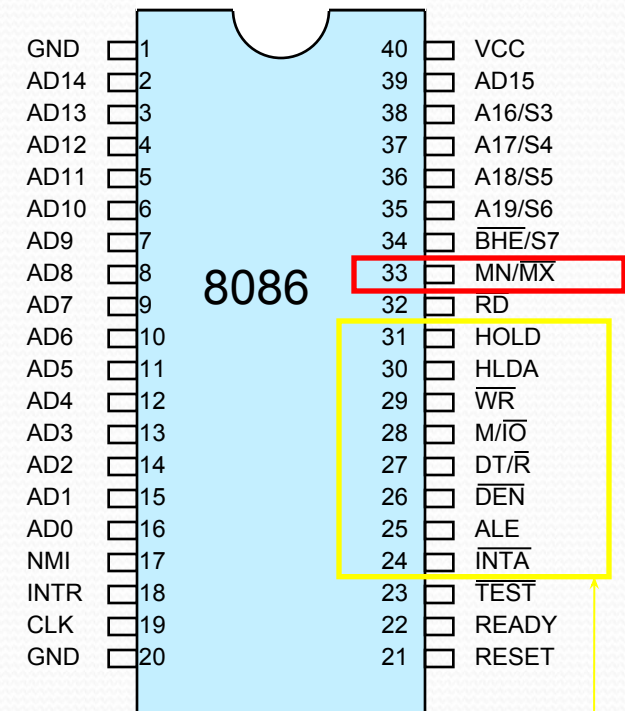
- 8086 microprocessors can be configured to work in either of the two modes: the minimum mode and the maximum mode

✓ Minimum mode:

- Pull $\overline{MN}/\overline{MX}$ to logic 1
- Typically smaller systems and contains a single microprocessor
- Cheaper since all control signals for memory and I/O are generated by the microprocessor.

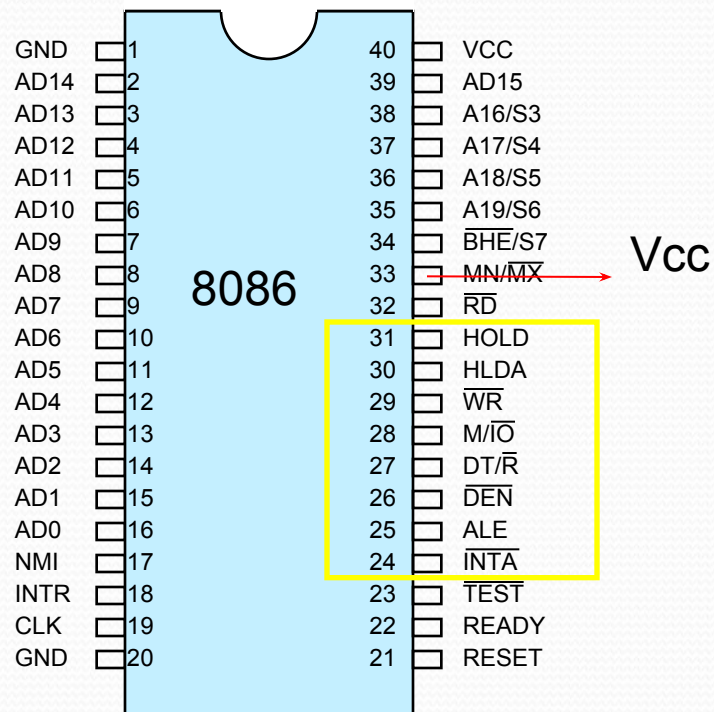
✓ Maximum mode

- Pull $\overline{MN}/\overline{MX}$ logic 0
- Larger systems with more than one processor (*designed to be used when a coprocessor (8087) exists in the system*)

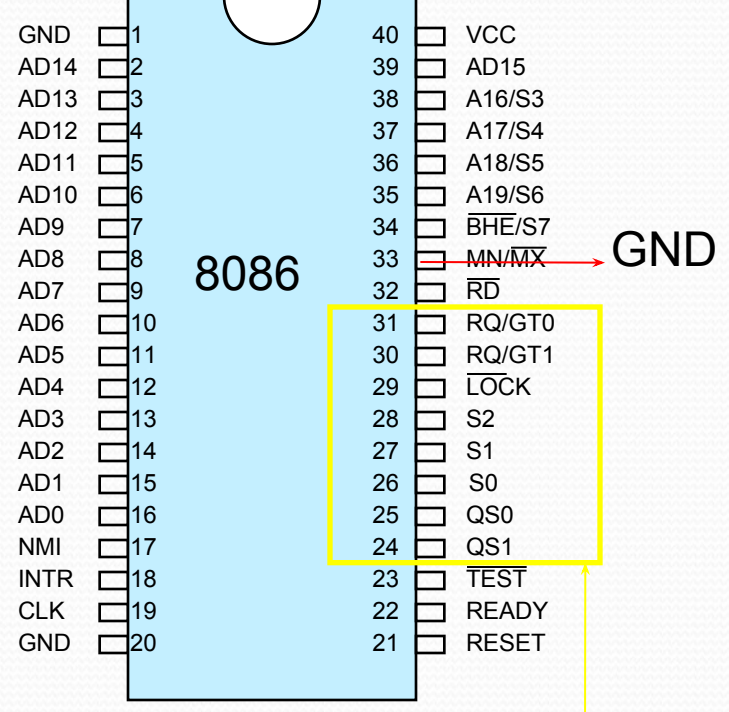


Lost Signals in
Max Mode

Minimum-mode and Maximum-mode Signals



Min Mode



Max Mode

AD₀ – AD₁₅

Pin 16-2, 39 (Bi-directional)

- These lines are multiplexed bi-directional address/data bus.
- During T₁, they carry lower order 16-bit address.
- In the remaining clock cycles, they carry 16-bit data.
- AD₀-AD₇ carry lower order byte of data.
- AD₈-AD₁₅ carry higher order byte of data.



$A_{19}/S_6, A_{18}/S_5, A_{17}/S_4, A_{16}/S_3$ Pin 35-38 (Unidirectional)

- These lines are multiplexed unidirectional address and status bus.
- During T_1 , they carry higher order 4-bit address.
- In the remaining clock cycles, they carry status signals.



$\overline{\text{BHE}} / \text{S}_7$

Pin 34 (Output)

- BHE stands for Bus High Enable.
- BHE signal is used to indicate the transfer of data over higher order data bus ($D_8 - D_{15}$).
- 8-bit I/O devices use this signal.
- It is multiplexed with status pin S_7 .



$\overline{RD}$ (Read)

Pin 32 (Output)

- It is a read signal used for read operation.
- It is an output signal.
- It is an active low signal.



READY

Pin 22 (Input)

- This is an acknowledgement signal from slower I/O devices or memory.
- It is an active high signal.
- When high, it indicates that the device is ready to transfer data.
- When low, then microprocessor is in wait state.



RESET

Pin 21 (Input)

- It is a system reset.
- It is an active high signal.
- When high, microprocessor enters into reset state and terminates the current activity.
- It must be active for at least four clock cycles to reset the microprocessor.



INTR

Pin 18 (Input)

- It is an interrupt request signal.
- It is active high.
- It is level triggered.



NMI

Pin 17 (Input)

- It is a non-maskable interrupt signal.
- It is an active high.
- It is an edge triggered interrupt.



TEST

Pin 23 (Input)

- It is used to test the status of math co-processor 8087.
- The BUSY pin of 8087 is connected to this pin of 8086.
- If low, execution continues else microprocessor is in wait state.



CLK

Pin 19 (Input)

- This clock input provides the basic timing for processor operation.
- It is symmetric square wave with 33% duty cycle.
- The range of frequency of different versions is 5 MHz, 8 MHz and 10 MHz.



V_{CC} and V_{SS}

Pin 40 and Pin 20 (Input)

- V_{CC} is power supply signal.
- +5V DC is supplied through this pin.
- V_{SS} is ground signal.



Pin Description for Minimum Mode

INTA

Pin 24 (Output)

- This is an interrupt acknowledge signal.
- When microprocessor receives INTR signal, it acknowledges the interrupt by generating this signal.
- It is an active low signal.



ALE

Pin 25 (Output)

- This is an Address Latch Enable signal.
- It indicates that valid address is available on bus $AD_0 - AD_{15}$.
- It is an active high signal and remains high during T_1 state.
- It is connected to enable pin of latch 8282.



DEN

Pin 26 (Output)

- This is a Data Enable signal.
- This signal is used to enable the transceiver 8286.
- Transceiver is used to separate the data from the address/data bus.
- It is an active low signal.



DT / $\bar{R}$

Pin 27 (Output)

- This is a Data Transmit/Receive signal.
- It decides the direction of data flow through the transceiver.
- When it is high, data is transmitted out.
- When it is low, data is received in.



M / $\overline{IO}$

Pin 28 (Output)

- This signal is issued by the microprocessor to distinguish memory access from I/O access.
- When it is high, memory is accessed.
- When it is low, I/O devices are accessed.



WR

Pin 29 (Output)

- It is a Write signal.
- It is used to write data in memory or output device depending on the status of M/IO signal.
- It is an active low signal.



HLDA

Pin 30 (Output)

- It is a Hold Acknowledge signal.
- It is issued after receiving the HOLD signal.
- It is an active high signal.



HOLD

Pin 31 (Input)

- When DMA controller needs to use address/data bus, it sends a request to the CPU through this pin.
- It is an active high signal.
- When microprocessor receives HOLD signal, it issues HLDA signal to the DMA controller.



Pin Description for Maximum Mode

QS₁ and QS₀

Pin 24 and 25 (Output)

- These pins provide the status of instruction queue.

QS ₁	QS ₀	Status
0	0	No operation
0	1	1 st byte of opcode from queue
1	0	Empty queue
1	1	Subsequent byte from queue



$\overline{S_0}, \overline{S_1}, \overline{S_2}$ Pin 26, 27, 28 (Output)

- These status signals indicate the operation being done by the microprocessor.
- This information is required by the Bus Controller 8288.
- Bus controller 8288 generates all memory and I/O control signals.



$\overline{S_0}, \overline{S_1}, \overline{S_2}$ Pin 26, 27, 28 (Output)

$\overline{S_2}$	$\overline{S_1}$	$\overline{S_0}$	Status
0	0	0	Interrupt Acknowledge
0	0	1	I/O Read
0	1	0	I/O Write
0	1	1	Halt
1	0	0	Opcode Fetch
1	0	1	Memory Read
1	1	0	Memory Write
1	1	1	Passive



LOCK

Pin 29 (Output)

- This signal indicates that other processors should not ask CPU to relinquish the system bus.
- When it goes low, all interrupts are masked and HOLD request is not granted.
- This pin is activated by using LOCK prefix on any instruction.



$\overline{RQ/GT_1}$ and $\overline{RQ/GT_0}$

Pin 30 and 31 (Bi-directional)

- These are Request/Grant pins.
- Other processors request the CPU through these lines to release the system bus.
- After receiving the request, CPU sends acknowledge signal on the same lines.
- $\overline{RQ/GT_0}$ has higher priority than $\overline{RQ/GT_1}$.



- $\overline{MN}/\overline{MX}$ is an input pin used to select one of these modes. When $\overline{MN}/\overline{MX}$ is HIGH, the 8086 operates in the minimum mode. When $\overline{MN}/\overline{MX}$ is LOW, 8086 is configured to support multiprocessor systems. In this case, the Intel 8288 bus controller is added to the 8086 to provide bus controls.

- Pins 2 through 16 and 39 (AD15 - AD0) are a 16-bit multiplexed address/data bus. During the first clock cycle AD0-AD15 are the low order 16 bits of address. The 8086 have a total of 20 address lines.

- The upper four lines (35 – 38) are multiplexed with the status signals for the 8086. These are the A16/S3, A17/S4, A18/S5, and A19/S6. During the first clock period of a bus cycle, the entire 20-bit address is available on these lines. During all other clock cycles for memory and I/O operations, AD15-AD0 contains the 16-bit data, and S3, S4, S5, and S6 become status lines. S3 and S4 lines are decoded as follows:

A17/S4 A16/S3 Function

0	0	Extra segment
0	1	Stack segment
1	0	Code or no segment
1	1	Data Segment

- Status bits S3 and S4 indicate the segment register that is being used to generate the address the address and bit S5 reflects the contents of the IF flag. S6 is always held at 0 and indicates that an 8086 is controlling the system bus.
- Pins 1 and 20 are grounded. Pin 17 is NMI. This is the Non Maskable Interrupt input activated by a leading edge. Pin 18 is INTR. INTR is the maskable interrupt input. Pin 19 is CLK is for supplying the clock signal that synchronizes the activity within the CPU.
- Pin 21 (RESET) is the system reset input signal. When the 8086 detects the positive going edge of a pulse on RESET, it stops all activities until the signal goes LOW. When the reset is low, the 8086 initializes as follows:

8086 Component	Content
Flags	Clear
IP	0000H
CS	FFFFH
DS	0000H
SS	0000H
ES	0000H
Queue	Empty

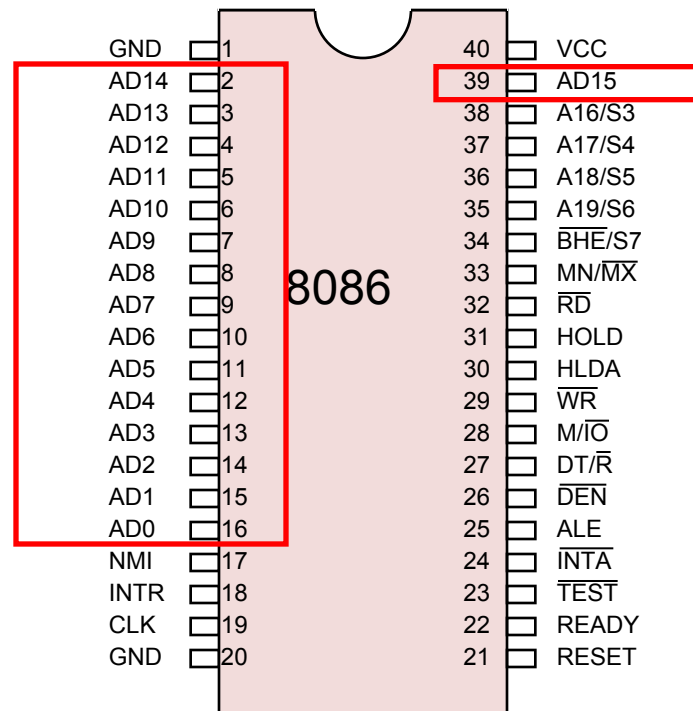
- Pin 22 (READY) is for inputting an acknowledge from a memory or I/O interface that input data will be put on the data bus or output data will be accepted from the data bus within the next clock cycle. In either case, the CPU and its bus control logic can complete the current bus cycle after the next clock cycle.
- Pin 23 TEST is an input pin and is only used by the WAIT instruction. The 8086 enter a wait state after execution of the WAIT instruction until a LOW is seen on the TEST pin.

- Pins 24 to 31 are mode dependent and are considered later.
- Pin 32 (RD) is LOW whenever the 8086 is reading data from memory or an I/O location.
- Pin 34 ($\overline{\text{BHE}}/\text{S7}$) is used as BHE (BUS High Enable) during the first clock cycle of an instruction execution. BHE can be used in conjunction with AD0 to select memory banks. During all other clock cycles $\overline{\text{BHE}}/\text{S7}$ is used as S7.

Operation	$\overline{\text{BHE}}/\text{AD0}$	Data pins used
Write/Read a word at an even Address	0 0	AD15 – AD0
Write/Read a byte at an even Address	1 0	AD7 – AD0
Write/Read a byte at an odd Address	0 1	AD15 – AD8
Write/Read a word at an odd Address	0 1 1 0	AD15 – AD8 AD7 - AD0

Multiplex of Data and Address Lines in 8086

- Address lines A0-A15 and Data lines D0-D15 are multiplexed in 8086. These lines are labelled as AD0-AD15.



Minimum-mode and Maximum-mode Systems

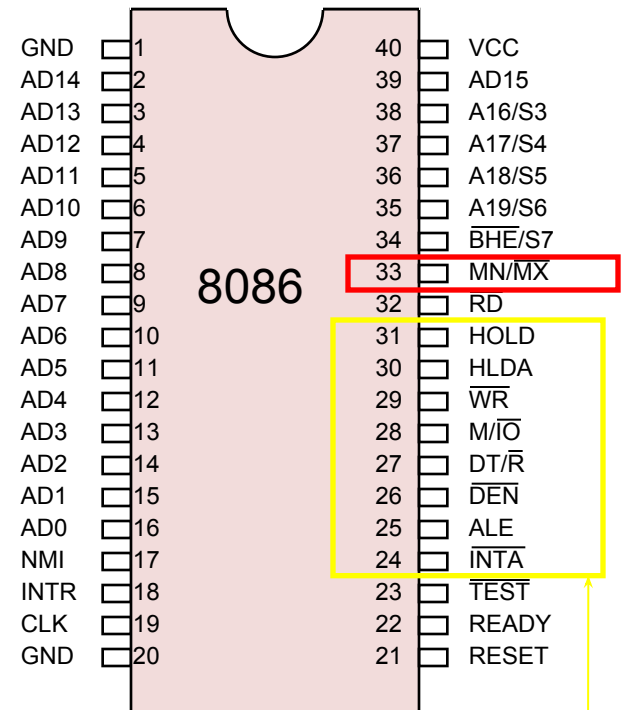
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Minimum mode:

- Pull MN/MX to logic 1
- Typically smaller systems and contains a single microprocessor
- Cheaper since all control signals for memory and I/O are generated by the microprocessor.

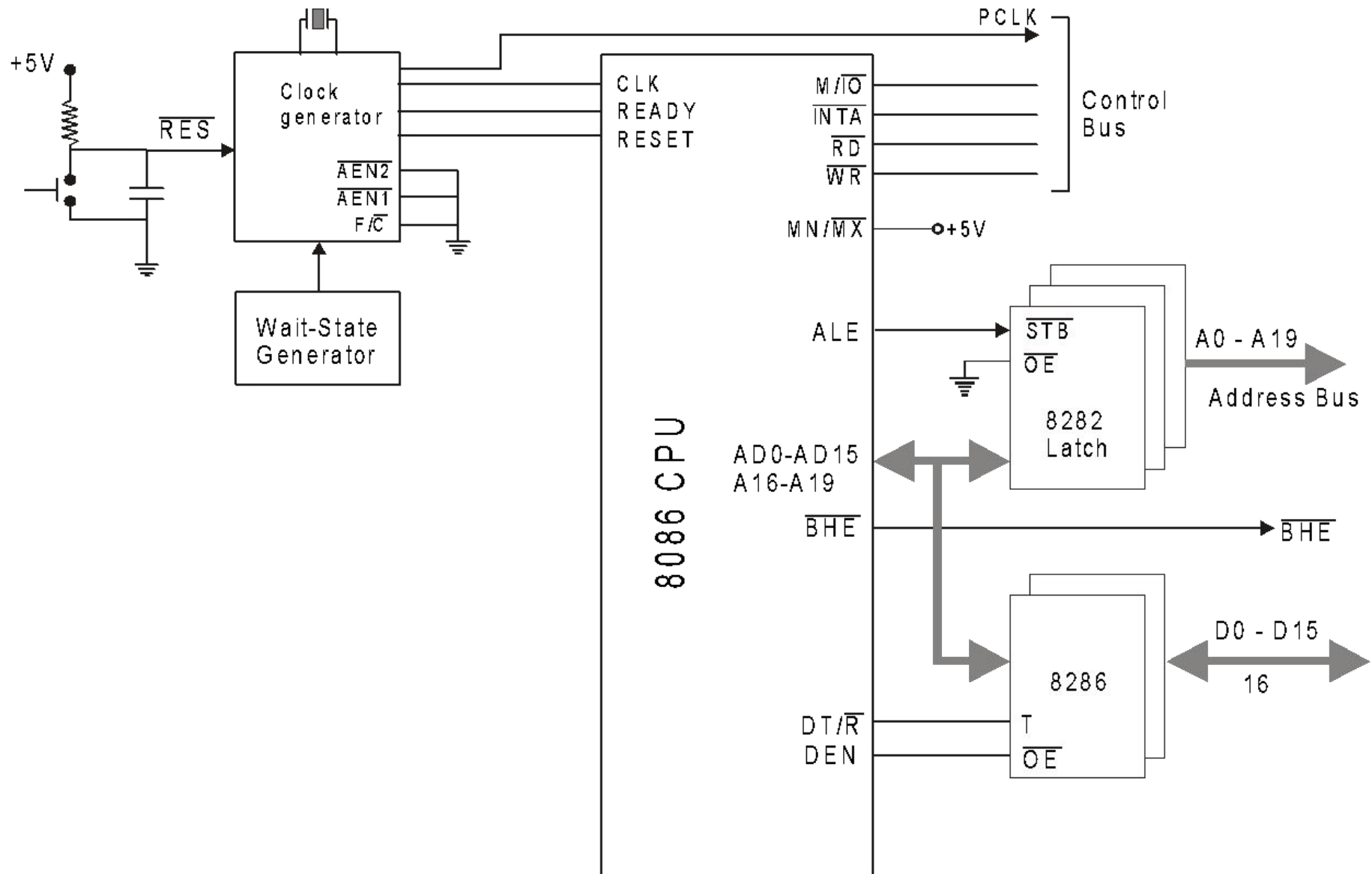
Maximum mode

- Pull MN/MX logic 0
- Larger systems with more than one processor (*designed to be used when a coprocessor (8087) exists in the system*)

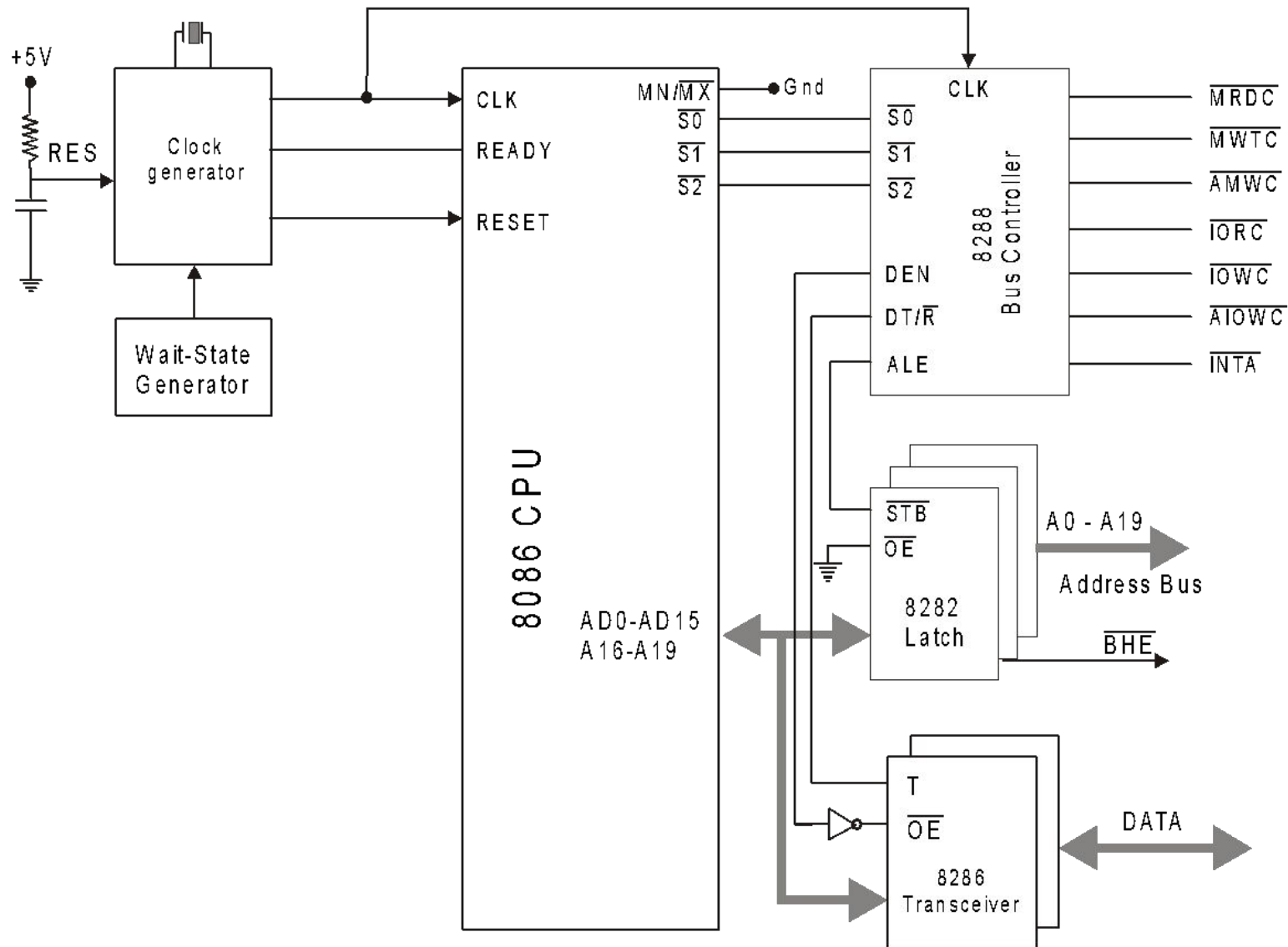


Lost Signals in
Max Mode

8086 System Minimum mode



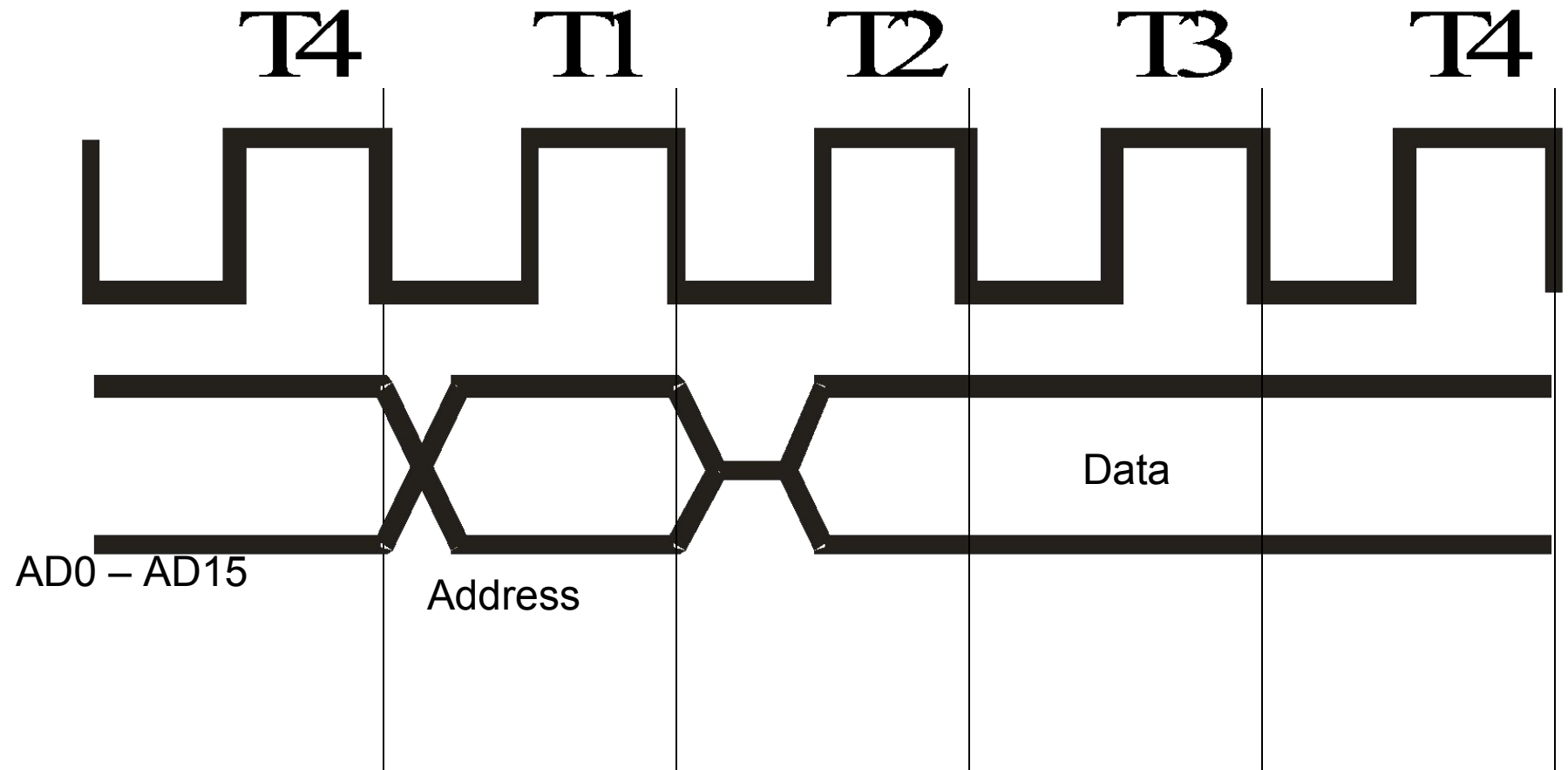
8086 System Maximum Mode



RESET Operation results

CPU component	Contents
Flags	Cleared
Instruction Pointer	0000H
CS	FFFFH
DS, SS and ES	0000H
Queue	Empty

AD0 - AD15: Address Data Bus



A17/S4, A16/S3 Address/Status

A17/S4	A16/S3	Function
0	0	Extra segment access
0	1	Stack segment access
1	0	Code segment access
1	1	Data segment access

A19/S6, A18/S5 Address/Status

A18/S5: The status of the interrupt enable flag bit is updated at the beginning of each cycle. The status of the flag is indicated through this pin

A19/S6: When Low, it indicates that 8086 is in control of the bus. During a "Hold acknowledge" clock period, the 8086 tri-states the S6 pin and thus allows another bus master to take control of the status bus.

S0, S1 and S2 Signals

S2	S1	S0	Characteristics
0	0	0	Interrupt acknowledge
0	0	1	Read I/O port
0	1	0	Write I/O port
0	1	1	Halt
1	0	0	Code access
1	0	1	Read memory
1	1	0	Write memory
1	1	1	Passive State

QS1 and QS2 Signals

QS1	QS1	Characteristics
0	0	No operation
0	1	First byte of opcode from queue
1	0	Empty the queue
1	1	Subsequent byte from queue



INTR (input)

Hardware Interrupt Request Pin

- INTR is used to request a hardware interrupt.
- It is recognized by the processor only when IF = 1, otherwise it is ignored (STI instruction sets this flag bit).
- The request on this line can be disabled (or masked) by making IF = 0 (use instruction CLI)
- If INTR becomes high and IF = 1, the 8086 enters an interrupt acknowledge cycle (INTA becomes active) after the current instruction has completed execution.

NMI (input) Non-Maskable Interrupt line

- The Non Maskable Interrupt input is similar to INTR except that the NMI interrupt does not check to see if the IF flag bit is at logic 1.
- This interrupt cannot be masked (or disabled) and no acknowledgment is required.
- It should be reserved for “catastrophic” events such as power failure or memory errors.

TEST (input)

- The TEST pin is an input that is tested by the WAIT instruction.
- If TEST is at logic 0, the WAIT instruction functions as a NOP.
- If TEST is at logic 1, then the WAIT instruction causes the 8086 to idle, until TEST input becomes a logic 0.
- This pin is normally driven by the 8087 co-processor (numeric coprocessor) .
- This prevents the CPU from accessing a memory result before the NDP has finished its calculation

Ready (input)

- This input is used to insert wait states into processor Bus Cycle.
- If the READY pin is placed at a logic 0 level, the microprocessor enters into wait states and remains idle.
- If the READY pin is placed at a logic 1 level, it has no effect on the operation of the processor.
- It is sampled at the end of the T2 clock pulse
- Usually driven by a slow memory device

HOLD (input)

- The HOLD input is used by DMA controller to request a Direct Memory Access (DMA) operation.
- If the HOLD signal is at logic 1, the microprocessor places its address, data and control bus at the high impedance state.
- If the HOLD pin is at logic 0, the microprocessor works normally.

HLDA (output)

Hold Acknowledge Output

- Hold acknowledge is made high to indicate to the DMA controller that the processor has entered hold state and it can take control over the system bus for DMA operation.